Supporting Information

How Far Can Structure-Based Machine Learning Predict Experimental Singlet–Triplet Gaps? An Honest Benchmark for TADF Emitter Triage

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S1. Computational details

S1.1 Semi-empirical calculations

Ground-state geometries were optimised with GFN2-xTB[1] (`xtb` 6.5.1, default convergence: 10^{-6} Ha SCF). Lowest singlet and triplet excitations were obtained with sTDA[2]. These semi-empirical excited states are used only to derive the natural transition orbital (NTO) descriptors of table S1; they are not used as a prediction target (the semi-empirical gap is circular with respect to such features; see main text).

S1.2 DFT reference calculations

Functional-dependence benchmarks (table S9) used B3LYP/6-31G(d) (Gaussian 16[3]) and CAM-B3LYP/def2-TZVP (ORCA 6.1.0[4]), in gas phase and toluene (CPCM).

S2. Natural transition orbital descriptors

NTOs[5] were obtained by singular-value decomposition of the transition density matrix; the dominant hole–electron pair ($\lambda_{\mathrm{NTO}} > 0.9$) defines the spatial descriptors. The 35 NTO descriptors—which require the sTDA excited-state calculation but exclude the S_1/T_1 energy scalars—are enumerated in table S1. For each of the lowest singlet (S_1) and triplet (T_1) states we computed the hole–electron spatial overlap S_{he}, the charge-transfer number CT, the donor/acceptor NTO weights λ_D/λ_A, the hole–electron separation Δr, and the localisation indices (hole-on-acceptor, particle-on-donor). State differences ($\Delta X = X_{S_1} - X_{T_1}$) and their magnitudes $|\Delta X|$ were added because the S_1–T_1 exchange splitting depends on how the spatial distributions *change* between states. The oscillator strength f_{S_1} and the log-transformed absorption proxies are the only intensity-related quantities; no energy scalar enters the feature set.

Table S1 The 35 NTO spatial descriptors used as the structure-derived feature set. X_{S_1} and X_{T_1} denote the quantity evaluated in the singlet and triplet state; $\Delta X = X_{S_1} - X_{T_1}$. Energy scalars ($E_{S_1}$, E_{T_1}, and the gap) are deliberately excluded.

Family	Descriptors
Hole–electron overlap (5)	$S_{\mathrm{he}}(S_1)$, $S_{\mathrm{he}}(T_1)$, ΔS_{he}, $\lvert\Delta S_{\mathrm{he}}\rvert$, `Char_diff_squared`
Charge-transfer number (4)	$\mathrm{CT}(S_1)$, $\mathrm{CT}(T_1)$, $\Delta\mathrm{CT}$, $\lvert\Delta\mathrm{CT}\rvert$
Donor weight (4)	$\lambda_D(S_1)$, $\lambda_D(T_1)$, $\Delta\lambda_D$, $\lvert\Delta\lambda_D\rvert$
Acceptor weight (4)	$\lambda_A(S_1)$, $\lambda_A(T_1)$, $\Delta\lambda_A$, $\lvert\Delta\lambda_A\rvert$
CT separation (4)	$\Delta r(S_1)$, $\Delta r(T_1)$, $\Delta(\Delta r)$, $\lvert\Delta(\Delta r)\rvert$
Localisation (4)	hole-on-A(S_1), particle-on-D(S_1), hole-on-A(T_1), particle-on-D(T_1)
NTO overlap composites (7)	$S^{\mathrm{sum}}_{\mathrm{NTO}}$, $S^{\mathrm{prod}}_{\mathrm{NTO}}$, $S^{\mathrm{ratio}}_{\mathrm{NTO}}$, ΔS_{NTO}, $\lvert\Delta S_{\mathrm{NTO}}\rvert$, NTO overlap($S_1$), NTO overlap($T_1$)
Intensity proxies (3)	f_{S_1}, $\log\lvert A_{S_1}\rvert$, $\log\lvert A_{T_1}\rvert$

S3. Machine learning details

S3.1 Target and dataset

The target is the experimental ΔE_{ST}, extracted from the literature and matched by compound name; multiple reports were reduced to the median. The benchmark contains 231 molecules spanning 212 Bemis–Murcko scaffolds[6], so scaffold-based splits place almost every molecule on a unique chemotype.

The corpus was assembled by a name-based join, which is a fragile step; we therefore state the rule and its accounting in full (table S2). Starting from 1 520 extracted literature rows, we kept rows whose property specifier contained "EST" (1 490), parsed the reported value (all 1 490 parsed), and retained values in the physical range $[-0.3, 1.5]$ eV (all 1 490). Each row's compound-name list was exploded and matched by exact string equality to the identifiers of molecules for which we had computed NTO descriptors, giving 363 name–value matches over 231 unique molecules. Of these, 102 molecules carried more than one literature report; we collapsed each to its median. The inter-report spread (standard deviation across reports for a given molecule) had a median of 0 eV but reached 0.564 eV in the worst case, which bounds the achievable accuracy from below. Every matched molecule had an RDKit-parseable SMILES (zero dropped), yielding the final set of 231 molecules over 212 scaffolds. The script `code/si_supplementary_data.py` regenerates this table.

Table S2 Provenance of the experimental ΔE_{ST} corpus: rows retained or dropped at each stage of the name-based join.

Stage	Count
Extracted literature rows	1520
Specifier contains "EST"	1490
Value parsed and finite	1490
Value in physical range $[-0.3, 1.5]$ eV	1490
Name–value matches to descriptor set	363
Unique molecules matched	231
of which with > 1 report	102
Dropped: no RDKit-parseable SMILES	0
Final benchmark molecules	**231**
Final Bemis–Murcko scaffolds	**212**

The corpus records no measurement medium: the solvent and atmosphere fields are empty for all 1 490 EST rows, so the target cannot be stratified by solvent. The label heterogeneity is instead quantified directly from the spread among independent reports of the same molecule—the empirical footprint of the uncontrolled solvent, host and method effects that dominate experimental ΔE_{ST} [7] (table S3). Across the 102 molecules with repeat reports, the inter-report range exceeds the model MAE (0.096 eV) for 19.6 % of molecules, and a single randomly chosen report sits 0.072 eV (RMS) from its molecule's median. This label noise is comparable to the model error, so it—not model capacity—sets the accuracy ceiling. Regenerated by `code/solvent_analysis.py`.

Table S3 Label heterogeneity of the experimental target, measured from within-molecule disagreement among independent literature reports (102 molecules with > 1 report). Range is max−min; SD is the standard deviation across a molecule's reports (the latter matches table S2).

Quantity (eV unless noted)	Value
Molecules with > 1 report	102
Inter-report range: mean / median	0.062 / 0.000
Inter-report range: 90th pct. / max	0.17 / 1.09
Fraction with range $>$ model MAE	19.6%
Inter-report SD: mean / max	0.036 / 0.564
Single report vs median (RMS)	0.072
Model MAE (reference)	0.096

S3.2 Models and evaluation

Model. Random forests [8] (400 trees, scikit-learn [9]) were compared against ridge regression, support-vector regression and a mean-value baseline; the random forest was best on every feature set. The forest used 400 estimators with no depth cap, $\sqrt{p}$ features per split, and a fixed seed; these defaults were not tuned per fold, so the reported scores carry no test-set selection bias.

Cross-validation. Performance is reported under Bemis–Murcko scaffold `GroupKFold` ($k = 5$): each fold holds out whole scaffolds, so no scaffold appears in both training and test, which removes the optimistic leakage of a random split on a set where 212 scaffolds cover 231 molecules.

Intervals and importance. 95 % confidence intervals come from 2 000 bootstrap resamples of the pooled out-of-fold predictions (table S4). Feature importance used exact TreeSHAP [10].

Table S4 Prediction of experimental ΔE_{ST} (random forest, scaffold cross-validation, 231 molecules); R^2 with 95 % bootstrap CI. The 2D-structure-only sets (no quantum chemistry) match the NTO set (which needs an sTDA run); structure-only R^2 intervals include zero.

Feature set	MAE (eV)	R^2 (95% CI)	ρ
Morgan FP (2D structure only)	0.091	0.27 (−0.19, 0.53)	0.31
RDKit desc. (2D structure only)	0.100	0.31 (−0.11, 0.51)	0.26
NTO spatial (needs sTDA)	0.096	0.25 (0.03, 0.45)	0.36
Energies + NTO (reference)	0.097	0.24	0.34
Mean-value baseline	0.107	0.00	
Raw sTDA gap	0.251	−2.22	0.24

Table S5 collects all accuracy and association metrics for the headline NTO model with their 95 % bootstrap confidence intervals, together with the power analysis for the rank-correlation test. Both Spearman ρ and Pearson r exclude zero, while the R^2 interval does not; the test for $\rho = 0.36$ is fully powered at $n = 231$ (only $n \approx 60$ is required for 80 % power), so the wide R^2 interval is a property of the variance-explained statistic under outlier influence rather than a sample-size limitation. Values regenerated by `code/r2_ci_power_analysis.py`.

Table S5 Headline NTO random-forest model: accuracy and association metrics with 95 % percentile-bootstrap CIs (5 000 resamples of the out-of-fold predictions, 231 molecules), and a Fisher-z power analysis for the correlation test (two-sided, $\alpha = 0.05$).

Metric	Point estimate	95% CI
MAE (eV)	0.096	(0.082, 0.109)
R^2	0.25	(0.03, 0.45)
Spearman ρ	0.36	(0.24, 0.47)
Pearson r	0.50	(0.26, 0.68)
Power analysis (correlation test, $\rho = 0.36$)		
Spearman ρ p-value	2.6×10^{-8}	
Achieved power at $n = 231$	≈ 1.0	
n for 80 % power	60	
n for 90 % power	79	

A scaffold-CV learning curve (fig. S1) tests whether the 231-molecule set is simply too small. The cross-validated MAE shows no systematic decrease beyond $n \approx 90$: it varies by 5 % across $n = 46$–231, within the fold-to-fold standard deviation (≈ 0.013 eV), while the training MAE stays near 0.035 eV. The persistent train–CV gap that does *not* close with more data is the signature of an irreducible label-noise floor rather than an under-sampled model; within this data regime, additional molecules of the same provenance would not lower the error. Regenerated by `code/learning_curve_analysis.py`.

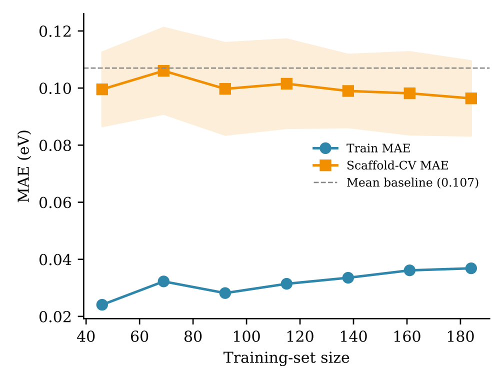

Figure S1 Scaffold cross-validation learning curve for the headline NTO random forest. Shaded band: ±1 fold standard deviation of the CV-MAE. The CV-MAE is flat beyond $n \approx 90$ and remains above the mean-value baseline gap, indicating a label-noise ceiling rather than a sample-size limit.

S3.3 Per-feature SHAP attributions

Table S6 lists the mean absolute TreeSHAP value of every NTO descriptor for the random forest trained on the 231-molecule set. Attribution concentrates on a handful of descriptors: the singlet hole–electron overlap $S_{he}(S_1)$ alone accounts for 29 % of the total attribution, and the top five descriptors ($S_{he}(S_1)$, S^{ratio}_{NTO}, f_{S_1}, $\Delta r(S_1)$, $S_{he}(T_1)$) for 60 %. The 12 lowest-ranked descriptors together contribute under 3 %; the two raw NTO-overlap and the squared-character features are effectively unused. This is consistent with the main-text finding that the gap is governed by a small number of donor–acceptor overlap quantities rather than by the full 35-dimensional descriptor vector.

Table S6 Per-feature importance (mean |SHAP|, in meV) of the 35 NTO descriptors for the random forest on the 231-molecule experimental set, ranked high to low. Regenerated by `code/si_supplementary_data.py`.

Descriptor	meV	Descriptor	meV
$S_{\mathrm{he}}(S_1)$	36.9	$\lvert\Delta\mathrm{CT}\rvert$	1.6
$S_{\mathrm{NTO}}^{\mathrm{ratio}}$	14.5	hole-on-A(T_1)	1.5
f_{S_1}	11.5	$\Delta r(T_1)$	1.5
$\Delta r(S_1)$	7.9	CT(S_1)	1.4
$S_{\mathrm{he}}(T_1)$	6.9	$\lvert\Delta\lambda_A\rvert$	1.3
$\log\lvert A_{T_1}\rvert$	6.9	ΔCT	1.2
$\Delta(\Delta r)$	4.8	particle-on-D(S_1)	0.7
ΔS_{he}	4.3	$\lambda_D(S_1)$	0.7
$\lvert\Delta(\Delta r)\rvert$	4.1	particle-on-D(T_1)	0.6
$\lvert\Delta S_{\mathrm{he}}\rvert$	4.0	$\lambda_A(T_1)$	0.6
$\lvert\Delta\lambda_D\rvert$	3.1	$\lambda_A(S_1)$	0.5
$\log\lvert A_{S_1}\rvert$	3.0	hole-on-A(S_1)	0.5
$\Delta\lambda_D$	2.6	ΔS_{NTO}	0.1
$\Delta\lambda_A$	2.0	$S_{\mathrm{NTO}}^{\mathrm{sum}}$	0.1
$\lambda_D(T_1)$	2.0	NTO overlap(T_1)	0.1
CT(T_1)	1.7	NTO overlap(S_1)	0.0
		$\lvert\Delta S_{\mathrm{NTO}}\rvert$	0.0
		$S_{\mathrm{NTO}}^{\mathrm{prod}}$	0.0
		`Char_diff_squared`	0.0

The interpretive value of the physics-informed descriptors is visible when their attribution is placed beside that of the equally accurate Morgan-fingerprint model (fig. S2). The ten highest-attribution NTO descriptors carry 79 % of the total and are all named physical quantities; the ten highest-attribution Morgan bits carry only 30 %, spread over 2 048 anonymous hashed substructure bits that resist mechanistic reading. Equal predictive accuracy thus buys interpretability only in the NTO representation. Regenerated by `code/shap_morgan_vs_nto.py`.

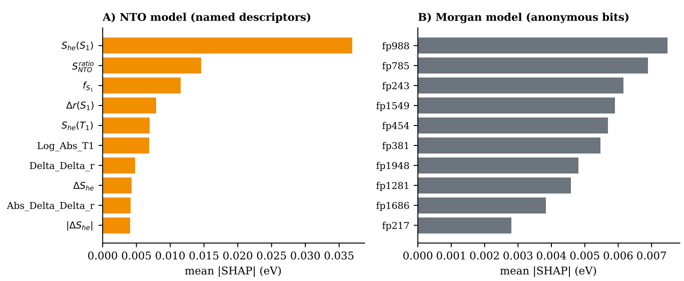

Figure S2 Top-ten SHAP attributions for the two equally accurate models. (A) NTO random forest: attribution concentrates on named donor–acceptor overlap and charge-transfer descriptors. (B) Morgan random forest: attribution is diffuse across anonymous fingerprint bits (fp$n$), which carry no direct chemical meaning.

To test whether this attribution pattern is a stable signal rather than an artefact of one particular fit, we refit the random forest on 200 bootstrap resamples of the 231-molecule set and recorded the top-five descriptors of each (table S7). The leading descriptor $S_{\mathrm{he}}(S_1)$ appears in the top five of 99.5 % of resamples; on average 3.7 of the five canonical descriptors recur per resample, and 64.5 % of resamples recover at least four of the five. The attribution is therefore reproducible at the top of the ranking, even though the model's ranking power on the gap itself is modest; we accordingly read these attributions as mechanistic hypotheses for experimental test, not validated causal claims. Regenerated by `code/shap_bootstrap_stability.py`.

Table S7 Bootstrap stability of the top-five SHAP descriptors: fraction of 200 resampled-and-refit models in which each canonical descriptor remains in the top five.

Descriptor	Top-5 frequency
$S_{\mathrm{he}}(S_1)$	99.5%
$S_{\mathrm{NTO}}^{\mathrm{ratio}}$	74.0%
f_{S_1}	72.0%
$\Delta r(S_1)$	68.0%
$S_{\mathrm{he}}(T_1)$	59.5%
Mean top-5 overlap	3.7 / 5
Resamples recovering ≥ 4	64.5%

S3.4 Scaffold-split characterisation

Because 212 Bemis–Murcko scaffolds cover 231 molecules, 92.9 % of scaffolds are singletons, and a reviewer may reasonably ask whether scaffold `GroupKFold` differs at all from a random split. It does, in the way that matters for leakage: by construction no scaffold appears in both training and test (zero shared groups in every fold), whereas a random split of the same data leaks 3 to 8 scaffolds per fold across the train/test boundary (table S8). The numerical effect on difficulty is nonetheless small—the mean nearest-neighbour Tanimoto similarity between a test molecule and its closest training molecule falls only from 0.615 (random) to 0.592 (scaffold)—so we do not claim the split makes the task dramatically harder; we claim it removes a specific leakage channel.

That the model learns chemical signal rather than scaffold-group artefacts is established by a label-permutation test: the target was shuffled 1 000 times and the scaffold-CV MAE recomputed under each permutation. The true MAE (0.095 6 eV) lies far in the left tail of the null distribution (mean 0.121 eV, SD 0.004 3 eV, 95 % interval 0.112 to 0.129), giving $p = 0.001$. Regenerated by `code/scaffold_split_analysis.py`.

Table S8 Scaffold `GroupKFold` versus a random 5-fold split on the same 231 molecules. Scaffold splitting removes train/test scaffold overlap; the label-permutation test confirms the fitted model is far from the no-signal null.

Quantity	Scaffold split	Random split
Shared scaffolds per fold	0 (all folds)	3–8
Test–train NN Tanimoto (mean)	0.592	0.615
Label-permutation test (1000 shuffles, scaffold CV)		
True CV-MAE (eV)	0.0956	
Null CV-MAE: mean (95% CI) (eV)	0.121 (0.112, 0.129)	
p-value (left tail)	0.001	

S3.5 Reproducibility

The scripts `code/finalize_model.py` and `code/rebuild_structural_model.py` regenerate table S4 and Fig. 1 of the main text from the deposited data.

S4. DFT level benchmark

Across 7 representative emitters in gas and toluene, B3LYP/6-31G(d) and CAM-B3LYP/def2-TZVP differ in ΔE_{ST} by a mean absolute 0.30 eV (up to 0.58 eV), with sign reversals of the relative ordering (e.g. DMAC-DPS). This functional dependence sets a physical ceiling on the meaning of any sub-0.1 eV accuracy claim (table S9).

Table S9 Functional dependence of ΔE_{ST} (eV): B3LYP/6-31G(d) vs CAM-B3LYP/def2-TZVP.

Molecule	Phase	B3LYP	CAM-B3LYP	Δ
4CzIPN	gas	0.192	0.361	+0.169
4CzIPN	toluene	0.206	0.335	+0.129
ACRFLCN	gas	0.007	0.590	+0.583
ACRFLCN	toluene	0.000	0.548	+0.548
ACRSA	gas	0.034	0.447	+0.413
ACRSA	toluene	0.000	0.372	+0.372
BACN	gas	0.524	0.991	+0.467
BACN	toluene	0.521	0.921	+0.400
DMAC-DPS	gas	0.489	0.188	−0.301
DMAC-DPS	toluene	0.444	0.207	−0.237
DMAC-TRZ	gas	0.098	0.261	+0.163
DMAC-TRZ	toluene	0.085	0.271	+0.186
PXZ-NAI	gas	0.245	0.355	+0.110
PXZ-NAI	toluene	0.289	0.360	+0.071

A higher-tier check confirms this functional dependence is not an artefact of the disparate B3LYP/CAM-B3LYP pair. Across six range-separated and hybrid functionals applied to 27 charge-transfer emitters[11], the singlet–triplet gap *estimate* ($E(\mathrm{S}_1^{\mathrm{ROKS}}) - E(\mathrm{S}_1^{\mathrm{UKS}}) \approx \Delta E_{\mathrm{ST}}/2$) disagrees per molecule by a median of 0.049 eV (75th percentile 0.10 eV, one outlier excluded). Even among modern functionals the estimate varies at the 0.05 eV to 0.10 eV level; no absolute gap or experimental agreement is claimed from these data. Extracted by `code/parse_roks_benchmark.py` from `data/theoexp_roks_energies.csv`.

The observed error floor is consistent with a root-sum-square combination of three *independently measured* noise sources (table S10). Treating them as uncorrelated gives an expected MAE floor of 0.128 eV, bracketing the observed random-forest MAE of 0.096 eV (95 % CI 0.082 to 0.109) and confirming the ceiling is set by the data, not the learner (fig. S3). Computed by `code/ceiling_decomposition.py`.

Table S10 Quantitative decomposition of the accuracy ceiling into three measured noise components. The root-sum-square floor is compared with the observed random-forest MAE and the mean-value baseline.

	Source	Magnitude (eV)
A	Inter-laboratory report scatter (RMS)	0.072
B	Vertical→adiabatic offset (MAE)	0.195
C	Functional dependence (mean spread)	0.296
	Root-sum-square floor	0.128
	Observed RF MAE	0.096
	Mean-value baseline	0.107

S4.1 Vertical versus adiabatic gap

The NTO descriptors are evaluated at the ground-state (vertical) geometry. To bound the error this introduces relative to a relaxed (adiabatic) treatment, we recomputed ΔE_{ST} for 14 representative emitters in gas and toluene at CAM-B3LYP/def2-TZVP (D3BJ, RIJCOSX) with excited-state geometry optimisation in ORCA, and compared with the vertical sTDA-xTB values (table S11). Over the 27 converged points the vertical gap deviates from the adiabatic reference by a mean absolute 0.195 eV (RMSE 0.245 eV, worst case 0.548 eV for 2CzTPE), but the deviation is a systematic offset, not random scatter: a linear fit of adiabatic on vertical gives slope 1.00 and intercept 0.07 eV ($R^2 = 0.70$), and

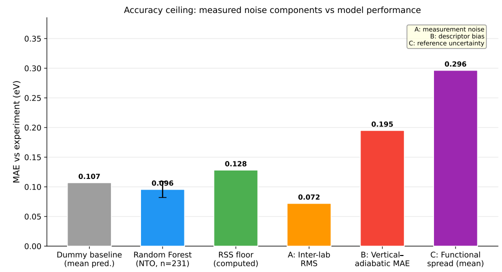

Figure S3 Accuracy-ceiling decomposition: the three measured noise components (A, B, C) and their root-sum-square floor against the observed random-forest MAE.

the rank correlation is preserved (Spearman $\rho = 0.71$). Geometry relaxation therefore shifts the absolute scale while leaving the ordering—on which the triage use depends—largely intact. Regenerated by `code/adiabatic_validation_analysis.py` from `data/vertical_vs_adiabatic_comparison.csv`.

Table S11 Vertical (sTDA-xTB, ground-state geometry) versus adiabatic (CAM-B3LYP/def2-TZVP, excited-state optimisation) ΔE_{ST} (eV) for 14 emitters. Δ = adiabatic − vertical. Summary: MAE 0.195 eV, slope 1.00, intercept 0.07 eV, $R^2 = 0.70$, $\rho = 0.71$.

Molecule	Phase	Vertical	Adiabatic	Δ
4CzIPN	gas	0.361	0.402	+0.041
4CzIPN	toluene	0.335	0.436	+0.101
DMAC-TRZ	gas	0.261	0.442	+0.181
DMAC-TRZ	toluene	0.271	0.638	+0.367
DMAC-DPS	gas	0.188	0.385	+0.197
DMAC-DPS	toluene	0.207	0.558	+0.351
PXZ-NAI	gas	0.355	0.243	−0.112
PXZ-NAI	toluene	0.360	0.491	+0.131
TPA-APy	gas	0.876	0.753	−0.123
TPA-APy	toluene	0.765	0.895	+0.130
BACN	gas	0.991	1.129	+0.138
BACN	toluene	0.921	1.063	+0.142
BMZ-TZ	gas	0.938	0.410	−0.528
BMZ-TZ	toluene	0.933	0.700	−0.233
2CzPN	gas	0.613	0.560	−0.053
2CzPN	toluene	0.621	0.652	+0.031
APPT-PXZ	gas	0.350	0.331	−0.019
2PXZP	gas	0.467	0.384	−0.083
2PXZP	toluene	0.485	0.639	+0.154
ACRSA	gas	0.447	0.204	−0.243
ACRSA	toluene	0.372	0.677	+0.305
ACRFLCN	gas	0.590	0.302	−0.288
ACRFLCN	toluene	0.548	0.576	+0.028
2CzTPE	gas	1.143	1.691	+0.548
2CzTPE	toluene	1.042	1.535	+0.493
BBPA	gas	1.521	1.641	+0.120
BBPA	toluene	1.433	1.562	+0.129

S4.2 Triplet manifold (T1–T2)

A small S_1–T_1 gap does not guarantee fast reverse intersystem crossing; the T_1–T_2 splitting and the wider triplet manifold matter too[12]. The same vertical CAM-B3LYP/def2-TZVP TD-DFT calculations that supplied the adiabatic benchmark also resolve ten triplet roots, so we report T_1 and T_2 directly (table S12). Across the 27 converged emitter–phase points the T_1–T_2 gap ranges from 0.018 eV (4CzIPN) to 1.50 eV (BBPA), mean 0.41 eV; 13 of 27 fall below 0.3 eV. The manifold therefore varies widely *among molecules with similar* ΔE_{ST} (e.g. DMAC-DPS and DMAC-TRZ have near-equal S_1–T_1 gaps but T_1–T_2 of 0.06 eV to 0.26 eV), which is exactly the information a scalar ΔE_{ST} triage cannot encode and a direction where the excited-state pipeline adds genuine value. Regenerated by `code/extract_triplet_manifold.py` from the `orca_package` outputs.

Table S12 Vertical CAM-B3LYP/def2-TZVP TDA energies (eV): S_1, T_1, T_2, the singlet–triplet gap $\Delta E_{\mathrm{ST}} = S_1 - T_1$, and the triplet-manifold gap T_1–T_2, for 14 emitters in gas and toluene.

Molecule	Phase	S_1	T_1	T_2	ΔE_{ST}	T_1–T_2
2CzPN	gas	3.652	3.039	3.464	0.613	0.425
2CzPN	toluene	3.632	3.011	3.459	0.621	0.448
2CzTPE	gas	3.496	2.353	2.767	1.143	0.414
2CzTPE	toluene	3.411	2.369	2.788	1.042	0.419
2PXZP	gas	3.292	2.825	3.080	0.467	0.255
2PXZP	toluene	3.343	2.858	3.139	0.485	0.281
4CzIPN	gas	3.377	3.016	3.050	0.361	0.034
4CzIPN	toluene	3.343	3.008	3.026	0.335	0.018
ACRFLCN	gas	3.597	3.007	3.548	0.590	0.541
ACRFLCN	toluene	3.546	2.998	3.536	0.548	0.538
ACRSA	gas	3.769	3.322	3.513	0.447	0.191
ACRSA	toluene	3.746	3.374	3.500	0.372	0.126
APPT-PXZ	gas	2.701	2.351	2.734	0.350	0.383
BACN	gas	3.713	2.722	3.533	0.991	0.811
BACN	toluene	3.629	2.708	3.503	0.921	0.795
BBPA	gas	3.553	2.032	3.525	1.521	1.493
BBPA	toluene	3.458	2.025	3.526	1.433	1.501
BMZ-TZ	gas	3.730	2.792	3.180	0.938	0.388
BMZ-TZ	toluene	3.896	2.963	3.142	0.933	0.179
DMAC-DPS	gas	3.637	3.449	3.508	0.188	0.059
DMAC-DPS	toluene	3.678	3.471	3.538	0.207	0.067
DMAC-TRZ	gas	3.509	3.248	3.504	0.261	0.256
DMAC-TRZ	toluene	3.533	3.262	3.509	0.271	0.247
PXZ-NAI	gas	2.688	2.333	2.836	0.355	0.503
PXZ-NAI	toluene	2.650	2.290	2.829	0.360	0.539
TPA-APy	gas	3.555	2.679	2.823	0.876	0.144
TPA-APy	toluene	3.425	2.660	2.795	0.765	0.135

To read the manifold across the whole corpus rather than the 14-molecule ORCA subset, we extracted T_1 and T_2 for all 231 emitters from the cheaper sTDA-xTB pipeline (`code/extract_t1_t2_rerun.py`; 231/231 converged, `data/t1_t2_full.csv`). The full-corpus T_1–T_2 gap has median 0.14 eV (mean 0.24 eV), and 74 % of emitters fall below the 0.3 eV region where a near-degenerate second triplet can open competing non-radiative channels (fig. S4). Most low-gap candidates therefore remain exposed to inefficient reverse intersystem crossing—a hazard a scalar S_1–T_1 triage cannot see.

This full-corpus manifold must be read *qualitatively only*. Benchmarking the sTDA-xTB T_1–T_2 gap against the CAM-B3LYP reference on the six overlapping molecules gives a mean absolute deviation of 0.237 eV (Spearman $\rho = 0.43$, $p = 0.40$; table S13), so the two methods agree on neither magnitude nor rank at a useful level. Quantitative ranking on RISC propensity still requires the higher-level calculation; the sTDA manifold serves only to flag that the triplet-manifold hazard is widespread, not to order candidates by it.

Table S13 sTDA-xTB versus CAM-B3LYP T_1–T_2 gap (eV) for the six molecules present in both sets. Independent methods; the comparison is genuine (fresh GFN2 geometries). Summary: MAE 0.237 eV, Spearman $\rho = 0.43$ ($p = 0.40$, $n = 6$).

Molecule	sTDA-xTB	CAM-B3LYP
2CzPN	0.169	0.425
2PXZP	0.004	0.255
4CzIPN	0.073	0.034
ACRSA	0.006	0.191
DMAC-DPS	0.011	0.059
PXZ-NAI	1.146	0.503

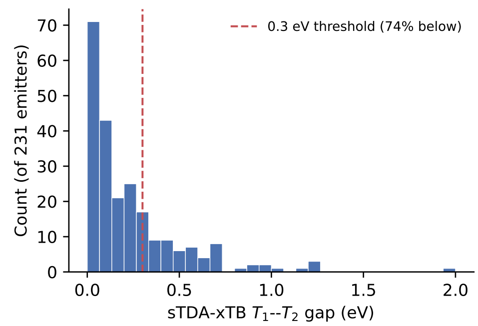

Figure S4 Distribution of the sTDA-xTB T_1–T_2 gap across all 231 emitters. 74 % fall below 0.3 eV (dashed line). Read qualitatively only (table S13).

S5. Heuristic library enumeration

An extended library of 22 194 donor–acceptor fragment combinations was enumerated from literature-validated donors (carbazole, phenoxazine, acridine) and acceptors (triazine, benzonitrile, sulfone) under simple constraints (MW < 900 Da, ≥ 2 aromatic rings, ≥ 1 N/O/S). Candidates were ranked by the rule-based TADF score defined in the main text. These are synthesis-planning leads only: they are not quantum-validated, and the heuristic score is not a calibrated ΔE_{ST} predictor. Property distributions of the enumerated set versus the top-scoring subset are shown in the main text.

To make the section actionable rather than decorative, we additionally rank the full library with the calibrated structure-only Morgan-FP model and attach a split-conformal 90 % prediction interval (half-width 0.18 eV, empirical coverage 0.89; table S14) to every prediction. The complete top-100 shortlist is deposited (`data/shortlist_top100.csv`). We stress the honest reading: the predicted gaps for the leading candidates (0.02 eV to 0.05 eV) sit well inside their own ± 0.18 eV intervals, so the list *prioritises* synthesis order; it does not certify any individual molecule. No device-level quantity is predicted. Generated by `code/build_shortlist.py`.

Table S14 Top 10 of the deposited 100-molecule shortlist: enumerated-library candidates ranked by predicted experimental ΔE_{ST} (structure-only Morgan-FP RF) with 90 % split-conformal prediction intervals (eV). SMILES truncated for display.

ID	SMILES (truncated)	Pred. (eV)	90% PI (eV)
MOL_016680	`O=C(C[C@@]1(O)C(=O)N(Cc2ccccc2Cl)c...`	0.023	[-0.15, 0.20]
MOL_016681	`O=C(C[C@]1(O)C(=O)N(Cc2ccccc2Cl)c2...`	0.023	[-0.15, 0.20]
MOL_002952	`O=c1c2ccccc2c(=O)c2c1ccc1c2[nH]c2c...`	0.034	[-0.14, 0.21]
MOL_000892	`Nc1cc2c(=O)c3cc(Cl)cc(Cl)c3[nH]c2c...`	0.035	[-0.14, 0.21]
MOL_000969	`O=C1c2ccccc2C(=O)c2c1c(Cl)cc1c(=O)...`	0.038	[-0.14, 0.22]
MOL_002522	`Cc1c2ccncc2c(C)c2c1[nH]c1c(Cl)cc(C...`	0.039	[-0.14, 0.22]
MOL_017220	`Cc1ccccc1CN1C(=O)[C@](O)(CC(=O)c2c...`	0.039	[-0.14, 0.22]
MOL_017221	`Cc1ccccc1CN1C(=O)[C@@](O)(CC(=O)c2...`	0.039	[-0.14, 0.22]
MOL_017226	`O=C(C[C@@]1(O)C(=O)N(CCc2ccccc2)c2...`	0.041	[-0.14, 0.22]
MOL_017407	`O=C(C[C@]1(O)C(=O)N(Cc2ccccc2)c2cc...`	0.041	[-0.14, 0.22]

S6. Robustness analyses

S6.1 Cluster cross-validation

Because 92.9 % of scaffolds are singletons, we built non-singleton groups by Butina clustering (Morgan Tanimoto distance) and ran cluster-grouped CV across a cutoff scan (table S15). Even at the coarsest grouping (70 clusters, largest 38 molecules), the MAE (0.099 eV) is statistically indistinguishable from scaffold-CV (0.096 eV) and random-CV (0.106 eV). Performance is therefore *robust to the validation scheme*: the headline number is not an artefact of singleton scaffolds. Regenerated by `code/cluster_cv_analysis.py`.

Table S15 Cluster-grouped CV of the headline NTO RF across Butina distance cutoffs, with scaffold and random CV for reference.

Grouping	#groups	largest	MAE (eV)	ρ
Butina 0.3	186	7	0.100	0.35
Butina 0.4	153	10	0.100	0.38
Butina 0.5	111	15	0.104	0.31
Butina 0.6	70	38	0.099	0.40
Scaffold (headline)	212	—	0.096	0.36
Random 5-fold	—	—	0.106	0.34

S6.2 Dimensionality and model capacity

Two checks address the charge that a 2 048-bit fingerprint overparameterises a 231-molecule set. First, shrinking the fingerprint to 1 024 or 512 bits leaves the MAE and the (negative) lower R^2 confidence bound essentially unchanged (table S16, top), so the negative R^2 tail is intrinsic to the low variance-explained on this target, not a width artefact; the lowest-dimensional regularised model (NTO + ElasticNet) brings the lower bound closest to zero (-0.03) but does not lift it. Second, no higher- or different-capacity learner beats the random forest under the same scaffold CV (table S16, bottom): gradient boosting, an MLP and an SVR are all worse. Model capacity is not the limiting factor. Restricting to the 208 molecules whose inter-report standard deviation is ≤ 0.05 eV (single-report molecules included, SD $= 0$) does not lower the MAE (0.106 eV), so the floor is not removed by discarding the noisiest labels either. Regenerated by `code/dimensionality_check.py` and `code/capacity_and_noise_ceiling.py`.

Table S16 Top: fingerprint-width sweep (RF) — MAE and 95 % bootstrap R^2 CI. Bottom: alternative learners under the same scaffold CV. None beats the RF.

Setting	MAE (eV)	R^2 (95% CI)
Morgan 512 (RF)	0.098	0.15 (−0.44, 0.46)
Morgan 1024 (RF)	0.096	0.24 (−0.25, 0.51)
Morgan 2048 (RF)	0.091	0.27 (−0.19, 0.53)
NTO + ElasticNet	0.102	0.10 (−0.03, 0.17)
RandomForest (headline)	0.096	—
HistGradientBoosting	0.105	—
MLP (100, 50)	0.169	—
SVR (RBF)	0.120	—
ElasticNet	0.102	—

S6.3 SHAP cross-check by permutation importance

An independent, model-agnostic permutation-importance analysis (held-out 30 %, 50 repeats) ranks the singlet hole–electron overlap $S_{\mathrm{he}}(S_1)$ *first*, in agreement with TreeSHAP; two of the SHAP top-five recur in the permutation top-five. The leading attribution is thus corroborated by two methods, while the lower-ranked tail is not — consistent with our restriction of mechanistic reading to the dominant descriptor. Regenerated by `code/permutation_importance_check.py`.

S6.4 Enrichment curve and calibrated operating points

Figure S5 reports precision (fraction of selected molecules with experimental gap ≤ 0.10 eV) against screening budget. Enrichment peaks at 1.37× over the 49 % base rate around the top 5–20% and does not improve at more aggressive cuts (the top 1% is two molecules). A split-conformal calibration gives a 90 % interval half-width of 0.178 eV (empirical coverage 0.892), so a laboratory can pick an operating point at a stated confidence. Regenerated by `code/enrichment_curve.py`.

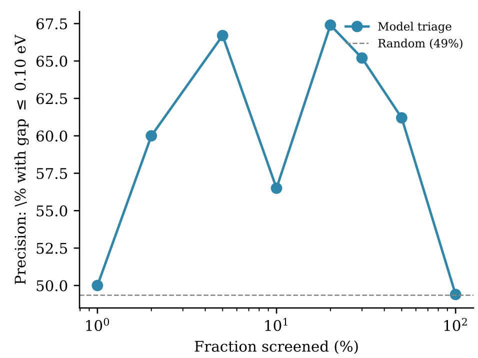

Figure S5 Triage precision versus fraction of the benchmark screened (log x). Dashed line: the 49 % random base rate. Enrichment is real but bounded at ≈ 1.3–$1.4\times$.

S6.5 Multi-criteria triage

A small S_1–T_1 gap is necessary but not sufficient for efficient delayed fluorescence: the emitter must also be optically bright and free of a near-degenerate second triplet. We therefore composed the predicted gap with two auxiliary structure-level filters—oscillator strength $f_{S_1} \geq 0.01$ (209/231 pass; `code/oscillator_strength_model.py`) and sTDA T_1–$T_2 > 0.10$ eV (table S13)—leaving 131 molecules that clear all three (`code/multi_criteria_triage.py`). Ranking within this pool raises the top-5 % precision for low-gap emitters from 0.58 (gap alone) to 0.71, i.e. enrichment 1.18→1.45×; the two curves converge by the top 50 %. The gain is real but small and rests on only a handful of molecules at the sharpest cut, so we read the multi-criteria filter not as a precision gain but as a physically motivated way to discard the 43 % of low-gap candidates that are dark or manifold-unsafe—a decision a scalar-gap triage cannot make. The T_1–T_2 term inherits the qualitative-only caveat of table S13.

S6.6 Dataset expansion stress-test

To test whether the benchmark is limited by corpus size, we resolved and featurised 624 additional donor–acceptor emitters from the experimental literature (name-to-structure resolution followed by the identical GFN2-xTB/sTDA natural-transition-orbital pipeline; `code/expand_dataset.py`, `code/featurize_nto_local.py`). After discarding 127 structures already present in the training set, 497 genuinely new molecules extend the corpus to 728 (table S17). Enlarging the set does *not* improve structure-only prediction: under the same scaffold CV the MAE rises from 0.098 eV to 0.124 eV and R^2 falls to −0.01 (the curated-set MAE of 0.098 eV reproduces the headline 0.096 eV within single-seed variation). Restricting the new molecules to those with replicate measurements (234 with ≥ 2 independent reports) recovers part of the loss (MAE 0.110 eV, R^2 0.05), placing one component of the degradation in label noise from automated literature extraction; the residual gap to the baseline reflects the greater chemical diversity of the larger set. A threefold expansion therefore does not lift the accuracy ceiling — it exposes it. Regenerated by `code/expansion_stress_test.py`.

Table S17 Scaffold-CV of the NTO random forest as the corpus is expanded and as new-label confidence is tightened. Expansion does not improve accuracy.

Corpus	n	MAE (eV)	R^2	ρ
Curated baseline	231	0.098	0.18	0.34
+ all new	728	0.124	−0.01	0.16
+ multi-report new (≥ 2)	465	0.110	0.05	0.24

S7. Data and code availability

All data (experimental corpus, descriptor tables, predictions), trained models, and analysis/figure scripts are openly available on Zenodo (DOI: 10.5281/zenodo.17436069) under a CC-BY-4.0 licence. Computations used `xtb` 6.7.1, `xtb4stda` with `stda` 1.6.3, Multiwfn 3.8, Gaussian 16, ORCA 6.1, Python 3.12, scikit-learn 1.8, RDKit 2026.03, and SHAP 0.50.

S8. Inverted singlet–triplet emitters (out of scope)

The benchmark is restricted to conventional TADF emitters with a positive gap ($\Delta E_{\mathrm{ST}} > 0$). The emerging inverted singlet–triplet (INVEST) class, which violates Hund's rule through double-excitation character[13], lies outside the reach of the present pipeline: GFN2-xTB enforces Aufbau ordering, and sTDA-xTB cannot reproduce the doubly-occupied LUMO character required for sign inversion, so a negative gap is structurally impossible for the random forest to predict from the current feature set. Screening the 25 482-record literature database (`code/ist_screening.py`) returns 66 reports of a reported inverted gap across 61 distinct compound names (25 DOIs), spanning experimental ΔE_{ST} from −0.43 eV to −0.01 eV (mean −0.11 eV; fig. S6, `data/ist_candidates.csv`). We do not classify their cores here because the database entries carry no curated SMILES. Of the 61, five also appear in our 231-molecule corpus; the random forest assigns a *positive* gap to all five (0.115 eV to 0.210 eV), i.e. the wrong sign in 5/5 cases. This is the expected falsification: INVEST screening requires ROKS- or EOM-CCSD-level descriptors[14] and is an explicit out-of-scope future axis, not a capability of a structure-only model trained on Aufbau-ordered features.

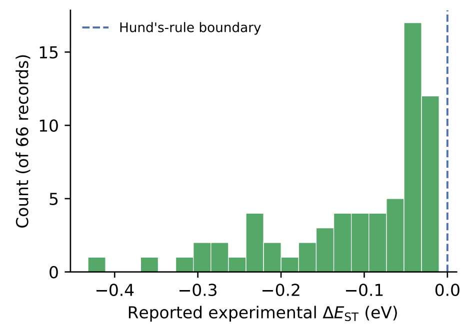

Figure S6 Reported experimental ΔE_{ST} for the 66 inverted-gap records identified in the literature database. All lie below the Hund's-rule boundary (dashed line); none is reachable by the Aufbau-ordered feature set.

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